

European Option Convergence Study

Neel Dev Sen

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Introduction

This convergence study is based on the 3 European Option pricers I have made in my other project called **Pricing Engines** (which you can find on my **GitHub**). The numerical methods that will be compared is the Binomial Tree, Trinomial Tree and the Finite Differences (using the Crank-Nicolson scheme)

1 Theoretical and Numerical Framework

This section will mainly just be stating the Time and Space complexities, the complete derivations of these are in this **document**.

1.1 Time Complexities

The time complexity of a Binomial Tree with N steps is:

$$\boxed{\mathcal{O}(N^2)} \tag{1.1}$$

The time complexity of a Trinomial Tree with N steps is also:

$$\boxed{\mathcal{O}(N^2)} \tag{1.2}$$

Finally the time complexity of the Finite Differences with a mesh of M space nodes and N time nodes is:

$$\boxed{\mathcal{O}(M \times N)} \tag{1.3}$$

Since I used the Thomas algorithm it makes it faster instead of $\mathcal{O}(M^3 \times N)$ In this experiment to keep this all similar so it is easier to compare, I define $M = N$ to make a square mesh such that the time complexity of Finite Differences is also:

$$\boxed{\mathcal{O}(N^2)} \tag{1.4}$$

All of these time complexities will be useful when I show the runtime-convergence graphs of these numerical methods

1.2 Space Complexities

The space complexity of a Binomial Tree with N steps is:

$$\boxed{\mathcal{O}(N)} \tag{1.5}$$

The space complexity of a Trinomial Tree with N steps is also:

$$\boxed{\mathcal{O}(N^2)} \quad (1.6)$$

Finally the time complexity of the Finite Differences with a mesh of M space nodes and N time nodes is:

$$\boxed{\mathcal{O}(M)} \quad (1.7)$$

Over here I don't store the entire matrix, instead I store the 3 diagonals as 3 separate vectors. Since again I am using $M = N$ this space complexity is equal to:

$$\boxed{\mathcal{O}(N)} \quad (1.8)$$

This will be useful for the section on memory.

1.3 Error Metrics

The **absolute error** is the main error metric that will be used to show convergence and it goes as shown.

$$\boxed{\text{Absolute Error} = |V_N - V_{CF}|} \quad (1.9)$$

where V_N is the value of the option given by the numerical method, and V_{CF} is the value of the option given by the closed form solution.

For the Arbitrage Error, I will just look at how far the options are from meeting the **Put-Call Parity** (the no-arbitrage condition) The Put Call Parity is as shown

$$C_t - P_t = S - Ke^{-r(T-t)} \quad (1.10)$$

The arbitrage error will be defined as:

$$\boxed{\text{Arbitrage Error} = |C_t - P_t - S + Ke^{-r(T-t)}|} \quad (1.11)$$

1.4 Data

The table below will be the data that is used in the convergence study.

Parameter	Black–Scholes	Black–76	Merton
Spot Price S_0	120.0	–	120.0
Futures Price F_0	–	115.0	–
Strike K	100.0	100.0	100.0
Risk-Free Rate r	0.04	0.04	0.04
Dividend Yield q	–	–	0.02
Volatility σ	0.25	0.25	0.25
Maturity T	2.0	2.0	2.0
Domain Multiplier	4.0	4.0	4.0

Table 1: Parameters used for the numerical convergence study.

2 Results: Black-Scholes

2.1 Black-Scholes Call

2.2 Black-Scholes Put

2.3 Arbitrage Error

2.4 Runtime

3 Results: Black-76

3.1 Black-76 Call

3.2 Black-76 Put

3.3 Arbitrage Error

3.4 Runtime

4 Results: Merton

4.1 Merton Call

4.2 Merton Put

4.3 Arbitrage Error

4.4 Runtime

5 Conclusion